

Pediatrics Massive Transfusion Protocol

Purpose:

- Resuscitation of children with severe bleeding due to trauma or other conditions may require massive transfusion. The pediatrics massive transfusion protocol (MTP) outlines the process for rapidly activating provision of blood products from Blood Bank to support the care of patients with active massive hemorrhage or if massive hemorrhage is anticipated. The massive transfusion protocol is initiated once the patient's main responsible provider on site (or assigned designate) has informed the Blood Transfusion Laboratory that a patient is experiencing massive blood loss. There is mandatory minimum data set required by Blood Bank on activation. Frequent monitoring of venous blood gas, Full Blood Count, coagulation studies, electrolytes and calcium are indicated.

Definition of Massive Transfusion:

- Massive transfusion can be defined as the replacement (or the anticipation of replacement) $\geq$ 1 blood volume within 24 hours or, 50% of blood volume within 4 hours, or, 10% of the blood volume over 10 minutes or 40 ml blood/kg.

Activating MTP:

Assigned Medical or nursing staff may activate the MTP if:

- Any child requiring more than 20mL/kg of packed red blood cells (PRBC) in 2 hours and anticipated ongoing blood loss;

OR

- Any child requiring more than 40mL/kg of PRBC in a 24-hour period with ongoing blood loss.

OR

- Any child requiring placement of >50% of total blood volume of blood products within 3 hours.

Note: An exception to the above criteria is planned cardiac bypass surgery. If blood loss or usage is greater than anticipated, the MTP may be initiated by the surgeon or anesthetist.

Estimated blood volume for different age group

Age	Estimated blood volume
Premature	90-100ml /kg
Neonate (0-4 kg)	85 mg/kg
Infant (5-9 kg)	85 mg/kg
Child (10-25kg)	75 ml/kg
Child (25-49 kg)	75 ml/kg
Obese child	65 ml/kg

Details required by the blood bank when activating MTP:

- MTP is activated by **calling blood bank (EXT 9711 & 9436)**
- When providing (verbal) information to Blood Bank, the following details are required:
- **Name and contact (phone/pager) of contact person.** (It is best to identify one person who will co-ordinate with Blood Bank. e.g. anesthetist/Surgeon in Operating Suite, emergency physician in the Emergency Department/ intensivist in PICU).
- **Name and medical record number (MRN) of the patient.**
- **Weight of patient.**
- **Location of patient and phone number of location.**

Note: Clinical team must notify Blood Bank if the patient location changes and details of the new responsible physician provider with full name of the new assigned delegate to communicate with the blood bank, new department, new phone number, and delivery information stating the patient's full name and MRN.

- **Initial blood results** e.g. full blood count (FBC), coagulation screen (coags), venous blood gas (VBG), cross match.
 - **Blood components transfused prior to arrival at hospital/activation of the Massive Transfusion protocol.**
 - **Urgency of need for blood products** e.g. immediate or over next 30 minutes. This will help determine what products will be dispatched
- .
- The script for the blood bank contact will read as follows, "My name is first name, last name and I am calling from the X department. We are calling a Pediatrics Massive transfusion Protocol on patient first name, last name, MRN. The approximate age and weight of the patient is X years and X kg. Based on the patient weight we will be ordering X units of PRBC + X units of FP. We need this to be delivered immediately/over next 30 minutes (decision as per clinical urgency)."
 - Blood Transfusion will send the PRBC as soon as they are ready, not waiting for the FP to thaw. The FP will be sent as soon as it is ready.
 - As soon as a batch of product is issued, the Blood Transfusion Laboratory will prepare subsequent products (same as the amount specified in the first order) for immediate transfusion and send, they will continue the process of blood product supply until the MTP is deactivated.

Role of Blood Bank:

- Supply initial requested units of Packed Red cells products (PRBCs)
- Once supplied, enquire the requesting medical officer if MTP need to be activated, once activated then to document the Medical officer's name.
- Notify the blood bank in-charge and dispense the required units as per MTP pack.
- If MTP is not activated and further PRBCs are requested within 24 hours then suggest MTP activation, document the health care provider Name, contact & location. Notify the on-call lab hematologist.

Time for blood product availability:

Product	Time Until Dispatch From Blood Bank
O negative PRBC, Platelets/ Group Specific if known blood group	Immediate
ABO specific PRBC (uncrossmatched)	Up to 5-10 minutes (Dispatch is immediate if group has already been verified by Blood Bank)
Crossmatched PRBC	45 minutes
FFP	30 Minutes
Cryoprecipitate	30 Minutes

MTP packs:

Each MTP pack contains PRBC, FFP, **Alternating units of platelets & cryoprecipitate**

Number of units in each pack will differ as per patient weight

1:1: ratio of PRBC to FFP

Weight	</ =10kg	11-25 kg	>26 kg
PRBC	20 ml/kg	2 units	4 units
FFP	20 ml /kg	2 units	4 units
Platelets	10 ml/kg	3 units	4 units
Cryoprecipitate	5 ml/kg	3 units	5-8 units

*PRBC = Packed Red Blood Cell

* FFP = Fresh Frozen Plasma

*If viscoelastic testing (e.g Rotational Thromboelastometry or Thromboelastography) is available, Clinicians may choose to utilize their expertise to incorporate it in the decision-making during a massive transfusion situation. It is often a more accurate assay of the entire hemostatic system including platelets function and fibrinolytic system (particularly useful in liver cirrhosis, disseminated intravascular coagulation or complicated coagulopathies) with rapid availability of result allowing timely intervention. This is being currently utilized in our institution in liver transplant surgeries.

Deactivation of MTP

- After the massive hemorrhage protocol is activated Blood Transfusion will prepare subsequent products for immediate transfusion hence MTP need to be canceled to stop the process of preparation.
- The clinician in charge of the patient (or delegate) **to notify Blood Bank of decreasing needs for products or termination of the massive transfusion protocol.**

Tranexamic Acid

The use of intravenous tranexamic acid in trauma has been established for adult patients (CRASH-2 trial) and administration is likely beneficial in children as well:

- Tranexamic acid at a dose of 15 mg/kg (dose range 10-20 mg/kg; max 1g) by slow intravenous infusion over 10-15 minutes; up to the first 3 hours after trauma.
- Follow up doses can be given at the decision of the treating team and can be as IV boluses or continuous infusion

Precautions:

- Avoid rapid infusion or push injection as it may cause hypotension.
- Tranexamic acid should be used with caution in patients with haematuria, renal haemorrhage and bleeding into other bodies cavities (e.g. pleural space, joints) as inhibition of fibrinolysis may result in retention of blood clots in those spaces.
- Dose/interval adjustments for subsequent doses are required in renal impairment

Additional Points to consider:

Use of crystalloids should be limited to avoid dilutional coagulopathy, hypothermia and hypertension.

Acidosis should be avoided. Treatment with sodium bicarbonate 2-3 mEq/kg may be considered if pH <7.2 or BD ≥ -6 and ventilation is not a problem.

For hyperkalemia, calcium chloride 10 mg/kg (max 1g) IV PRN can be used if central venous access is available. If central access is not available, calcium gluconate 30-50 mg/kg (max 1g) should be used.

Resuscitative end points (Target Values)

Haemoglobin >7 g/dl
Platelets >50 x 10 ⁹ (>100 if neurological injury or major cardiac/CNS surgery)
APTT <40s Fibrinogen >1 -1.5 Pt < 20 s
Core temperature > 36
Ph > 7.3

The algorithm outlining the process for handling a massive hemorrhage



Written By:

Dr Alyaa Al Mughairy (Pediatrics Hematology Oncology Department) & Dr Samiuddin (Pediatric Intensive care Department)

Reviewed By: Laboratory of Blood Transfusion Service

Appendix 1. Patient groups requiring irradiated or CMV negative blood products

The following groups of patients should receive irradiated red cells and platelets:

- All patients with Hodgkins lymphoma
- All patients treated with regimes including the following drugs: purine analogues (fludarabine, cladribine, deoxycoformycin, clofarabine, nelarabine, bendamustine),
- ATG, alemtuzumab (Campath)
- All recipients of allogeneic or autologous bone marrow or peripheral blood stem cell transplant
- All donors of bone marrow (BM) or peripheral blood stem cells (PBSC) from 7 days before harvest and during harvest
- All patients undergoing BM or PBSC harvest for later autologous transplant-from 7 days before harvest and during harvest
- Neonates who have previously received blood products in utero (required until 6 months after expected date of delivery)
- Children with severe T lymphocyte immunodeficiency syndromes (eg. SCID, CID, 22q11, Wiskott Aldrich syndrome)